Questions

Q1.

Figure 10 is a graph of intensity against wavelength for the electromagnetic radiation emitted by a halogen lamp.

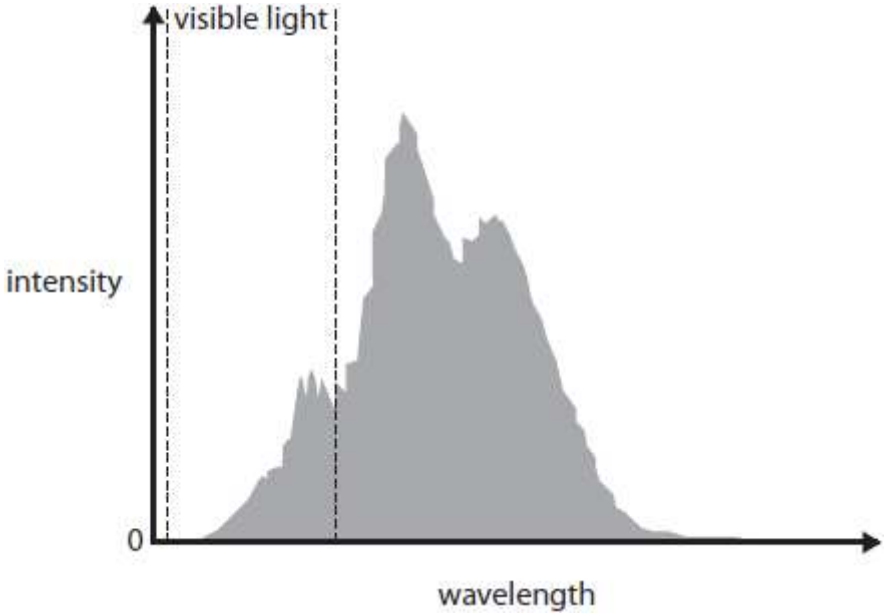

Figure 10

Describe how the intensity of the radiation varies with wavelength in Figure 10.

(2)

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(Total for question = 2 marks)

Q2.

This question is about different types of electromagnetic radiation.

Figure 9 shows the relative sensitivity, inside a dark cave, of an insect eye and a human eye.

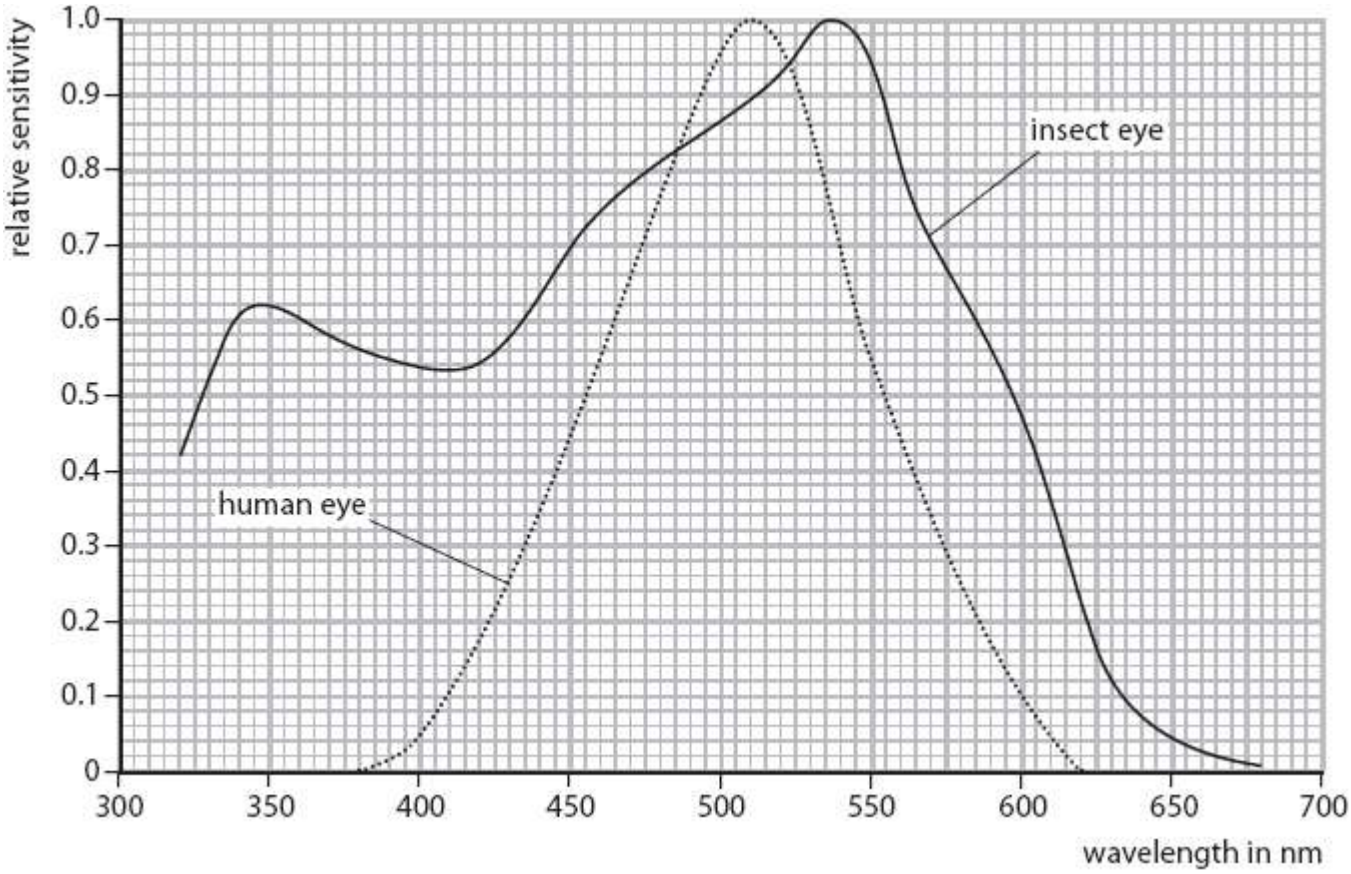

Figure 9

Using information from Figure 9, compare the relative sensitivities, inside the cave, of the insect eye with the human eye.

(4)

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(Total for question = 4 marks)

Q3.

(a) A specific electromagnetic wave has a frequency greater than visible light.
The wavelength of this wave is longer than that of X-rays.

Complete the sentence by putting a cross (☒) in the box next to your answer.

This electromagnetic wave is

(1)

☐ **A** a gamma wave

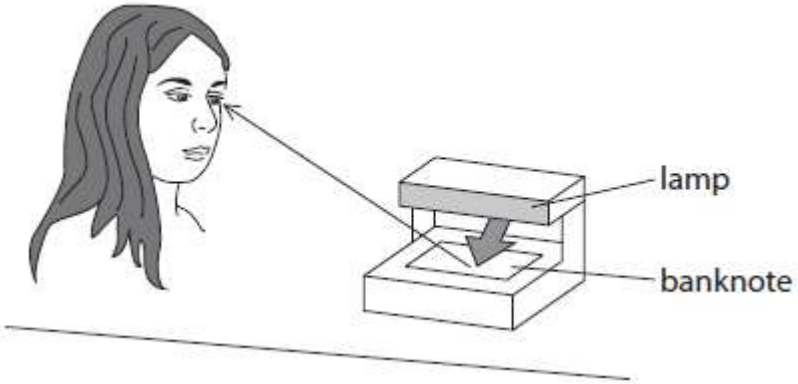
☐ **B** an infrared wave

☐ **C** a microwave

☐ **D** an ultraviolet wave

(b) The picture shows a woman checking that a banknote is genuine.

She is using a lamp which emits a radiation which is part of the electromagnetic spectrum.



Explain how **two** different electromagnetic radiations enable the woman to check the banknote.

(2)

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Q4.

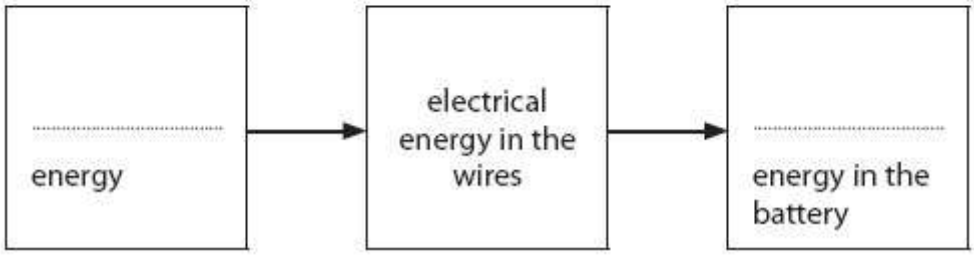
A student uses a solar powered battery charger to charge some batteries.

(a) The diagram is an energy transfer diagram for a battery being charged.

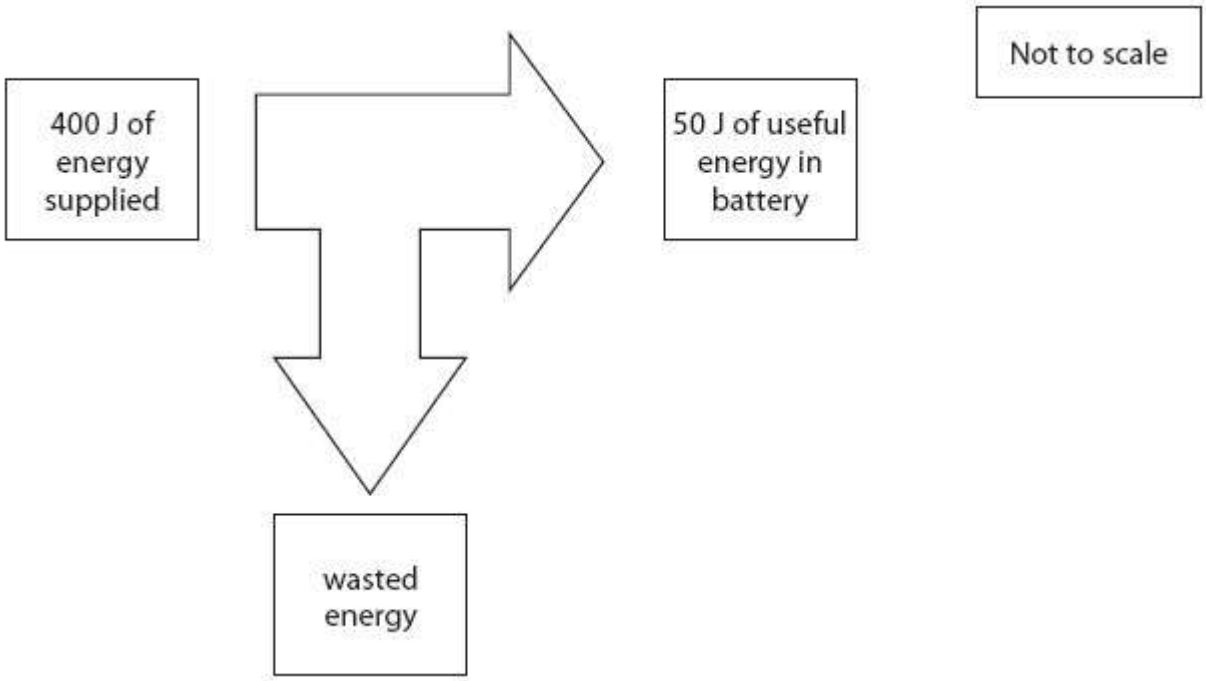
Use words from the box to complete the energy transfer diagram.

light	kinetic	sound
electrical	chemical	

Energy transfer diagram



(b) The diagram shows how much energy is usefully transferred by the battery charger.



(i) Calculate the amount of wasted energy.

(1)

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(ii) Calculate the efficiency of the battery charger.

(2)

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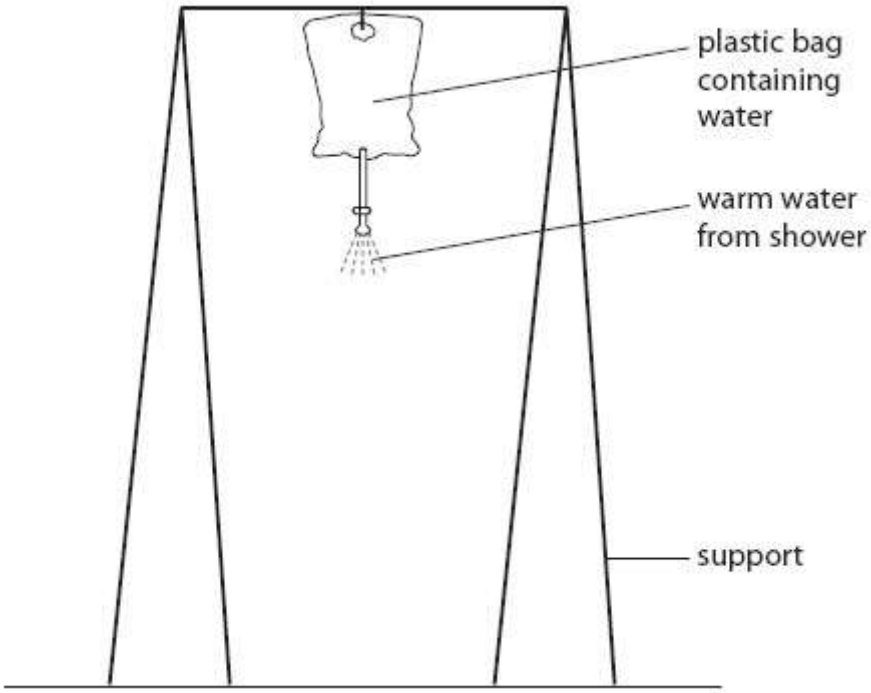
(c) The following arrangement is used as a solar powered shower.



The bag is left out in the sunlight during the day.

(i) Explain what colour the bag should be to heat the water to the highest temperature.

(2)

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(ii) On a sunny day the bag is filled with cold water.
Explain why the temperature of the water increases and then stays constant.

(3)

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(Total for Question = 10 marks)

Q5.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A student is investigating the refraction of light.

Figure 5 shows part of the apparatus and the angles to be measured.

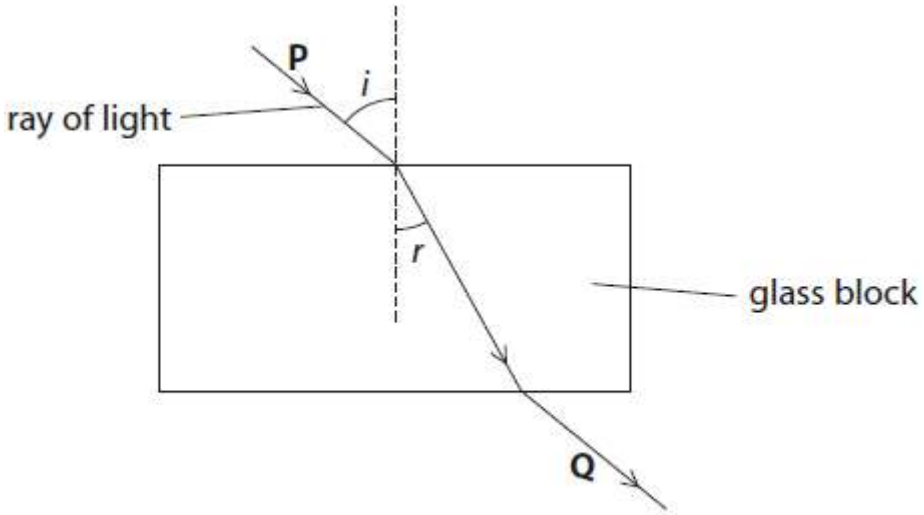


Figure 5

In Figure 5, the frequency of the light remains the same in glass as in air.

Which row of the table describes what happens to the wave velocity and to the wavelength of the light when the light travels from air to glass?

(1)

		wave velocity	wavelength
☐	**A**	decreases	decreases
☐	**B**	decreases	increases
☐	**C**	increases	decreases
☐	**D**	increases	increases

(Total for question = 1 mark)

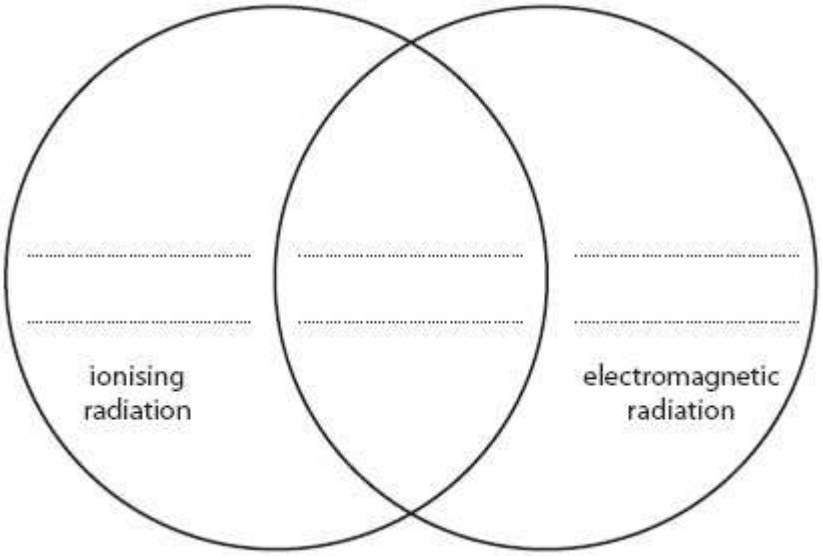
Q6.

(a) Skin cancer can be caused by radiation from the Sun.

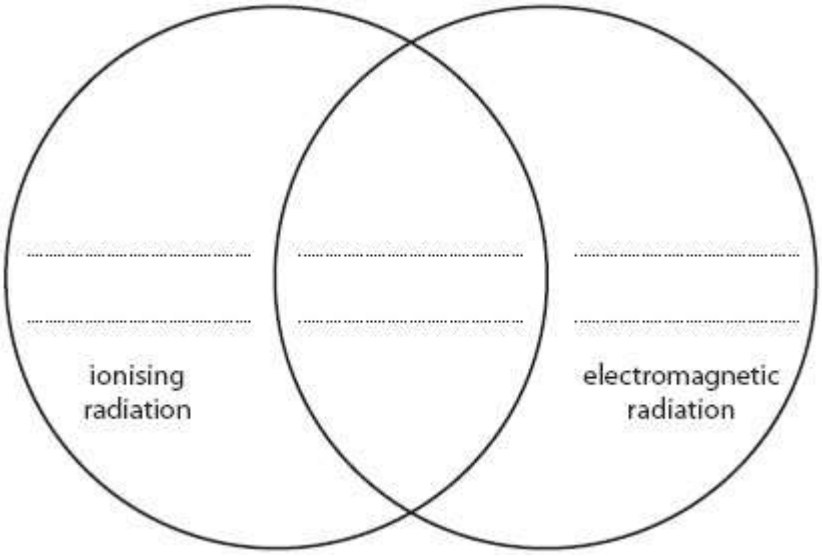
Complete the sentence by putting a cross (■) in the box next to your answer.

The radiation that causes skin cancer is

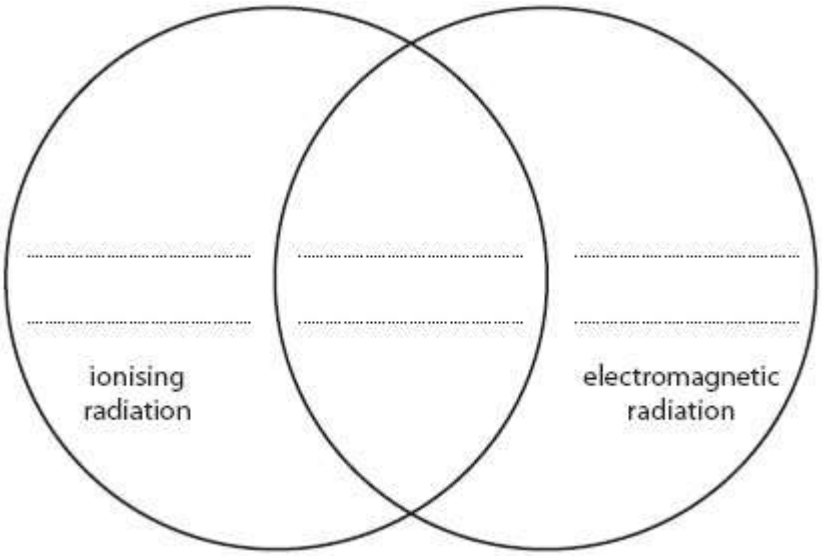
(1)



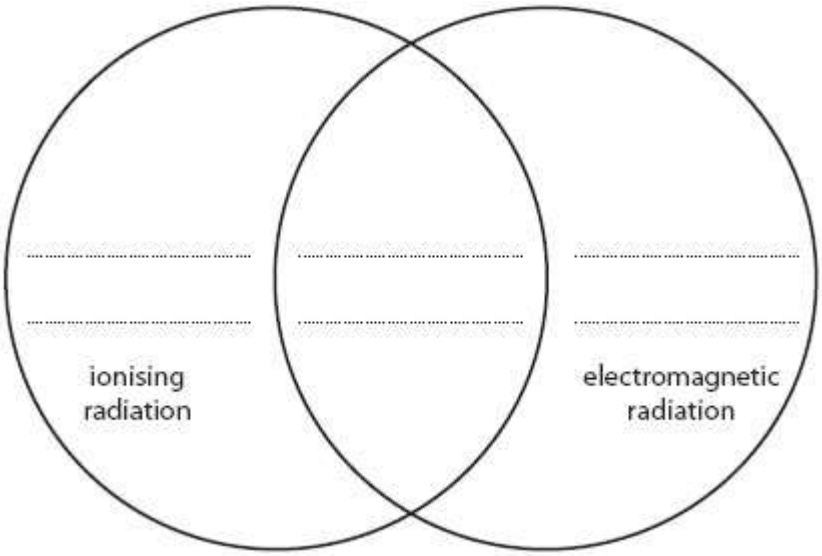
A ultraviolet radiation



B radio waves



C microwaves



D infrared radiation

(b) The word box contains the names of three types of radiation.

gamma rays	infrared radiation	alpha particles

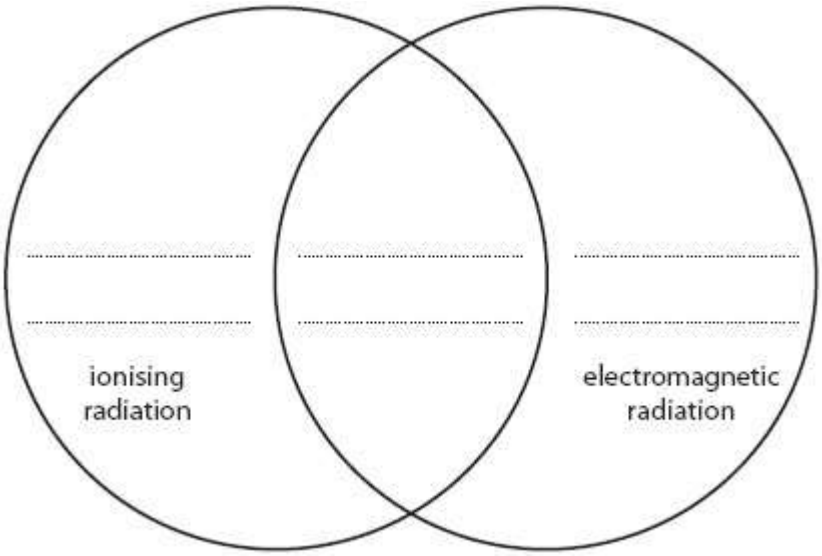
Use this diagram to classify the three types of radiation given in the word box.
Write the name of the radiation in the correct section of the diagram.

(2)

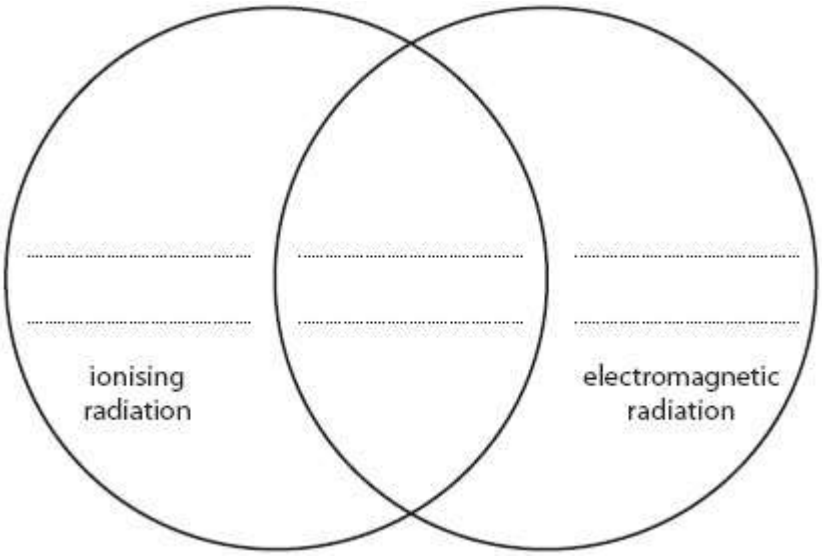
(c) Which of these is correct for all electromagnetic waves in a vacuum?

Put a cross (■) in the box next to your answer.

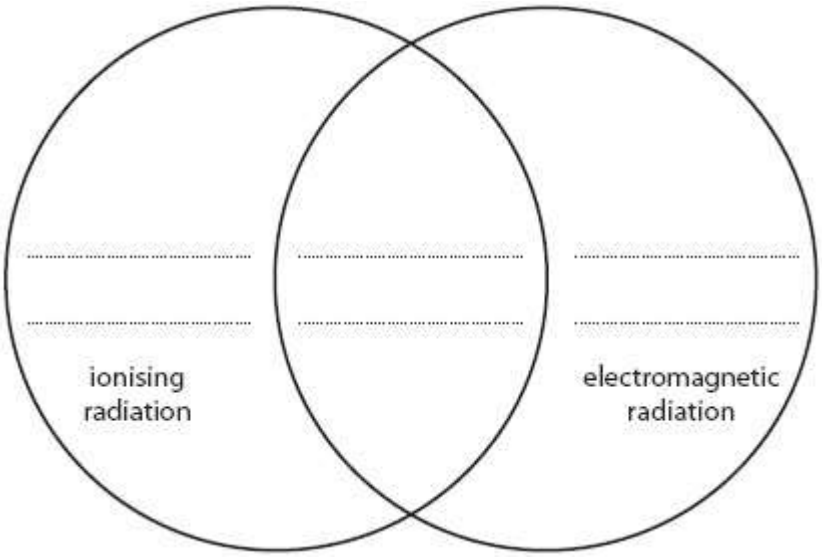
(1)



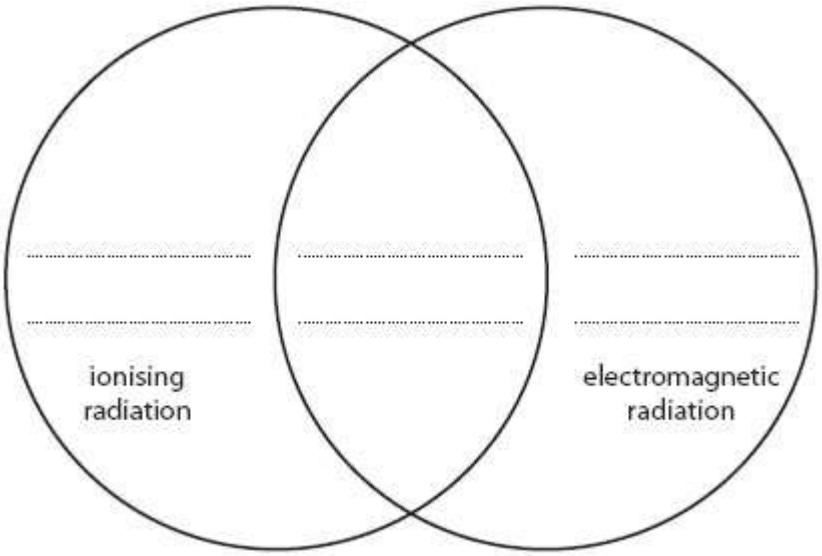
A they have the same frequency



B they have the same wavelength



C they are transverse waves



D they are longitudinal waves

(d) Describe a use of gamma radiation.

(2)

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*(e) Herschel and Ritter carried out experiments that contributed to the discovery of infrared and ultraviolet radiation.

Explain how the results of the experiments carried out by Herschel and Ritter led to these discoveries.

(6)

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(Total for Question is 12 marks)

Q7.

Figure 11 is a graph of temperature against time for a halogen lamp for the first 120 ms after it has been switched on.

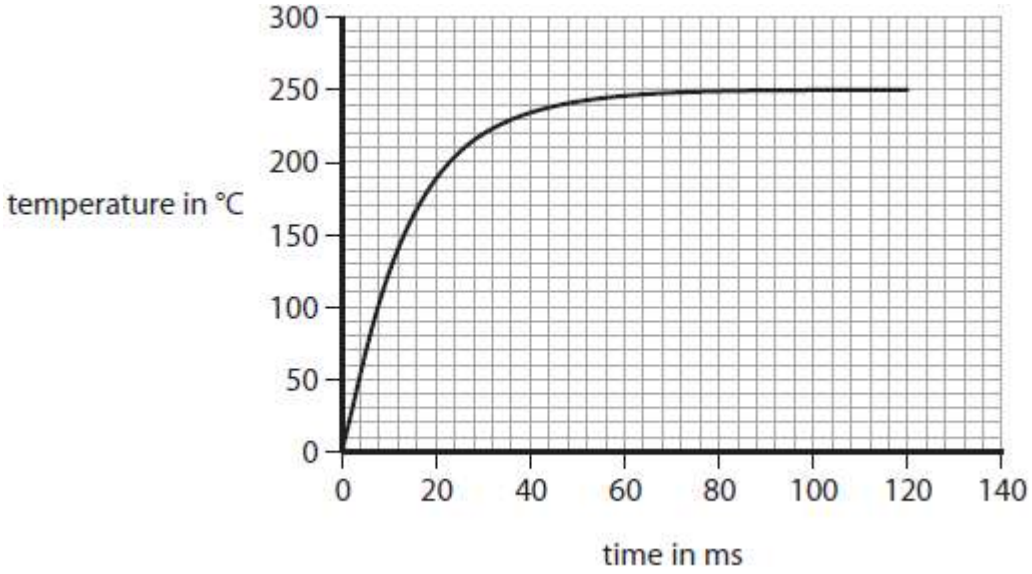

Figure 11

(i) Calculate the gradient of the graph at a time of 30 ms.
State the unit.

(4)

gradient = ... unit .....

(ii) Explain why the temperature of the lamp rises and then remains at a constant value.

(3)

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(Total for question = 7 marks)

Q8.

A student investigates how the angle of refraction varies with the angle of incidence when light enters a glass block.

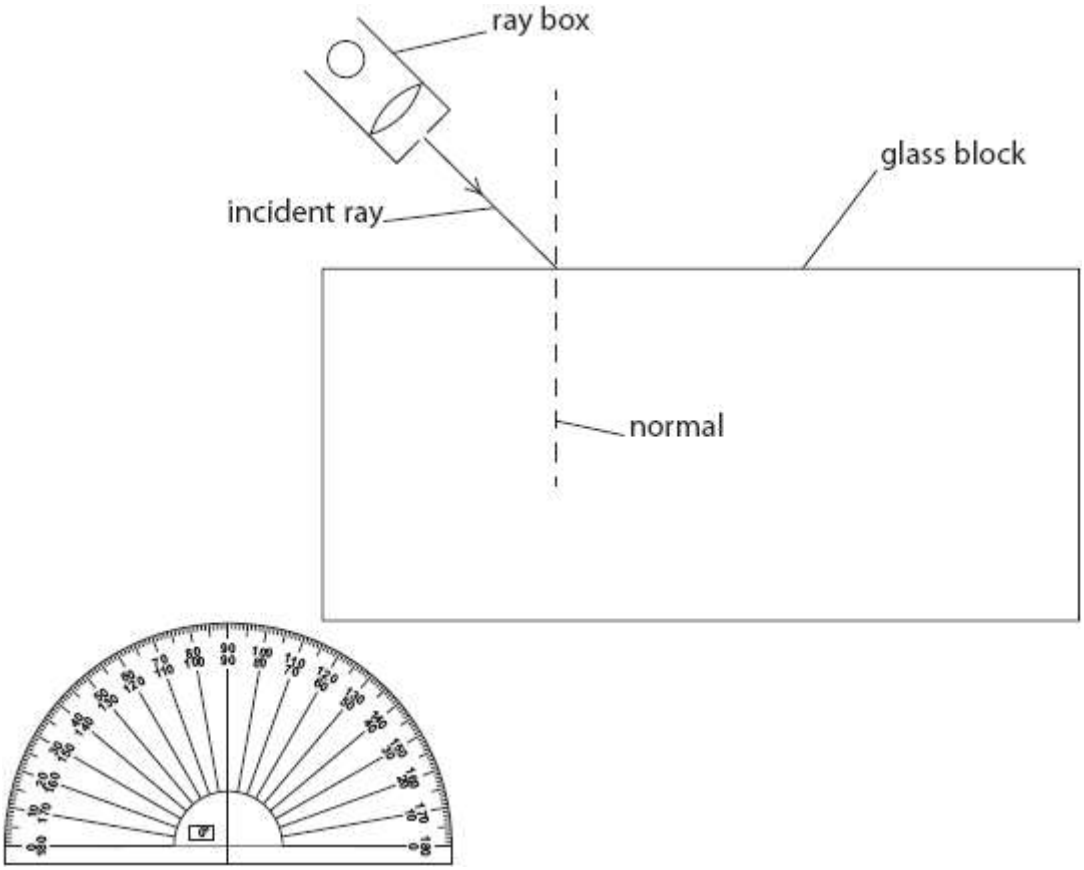

Figure 1

The student's results are shown in Figure 2.

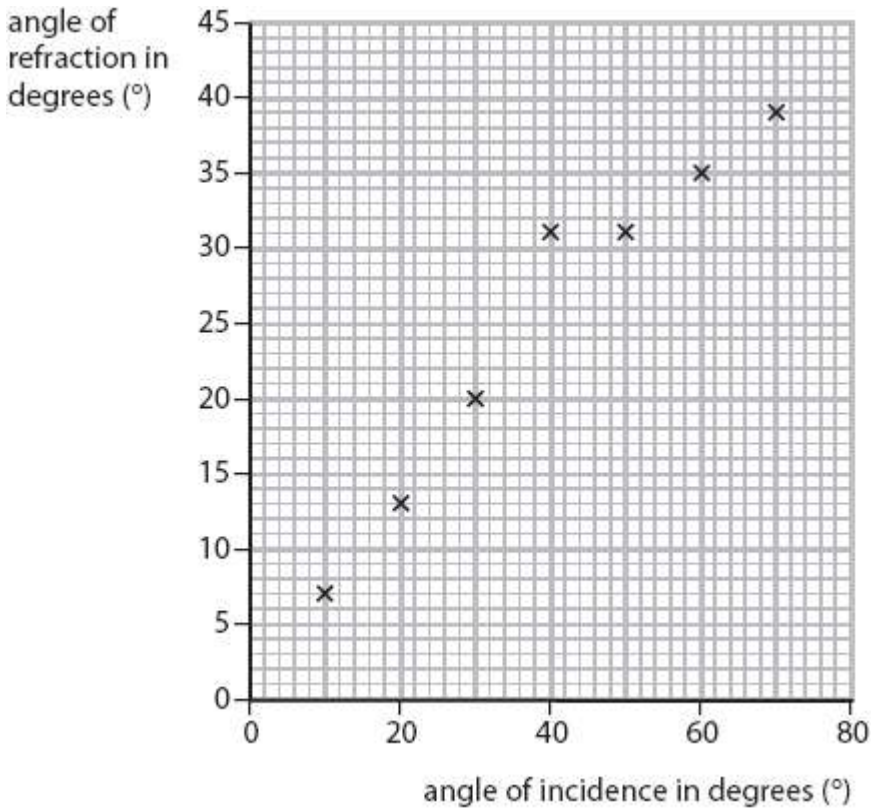

Figure 2

Explain whether any of the readings are anomalous.

(2)

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(Total for question = 2 marks)

Q9.

Some sunglasses have photochromic lenses.

Photochromic lenses are clear when the lenses are indoors but they darken in bright sunlight to reduce the effects of the sunlight.

Photochromic lenses react to ultraviolet light.

Suggest a benefit of making the lenses go dark with ultraviolet light.

(1)

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(Total for question = 1 mark)

Q10.

Figure 10 shows a ray diagram of the **virtual image** produced by a diverging lens.

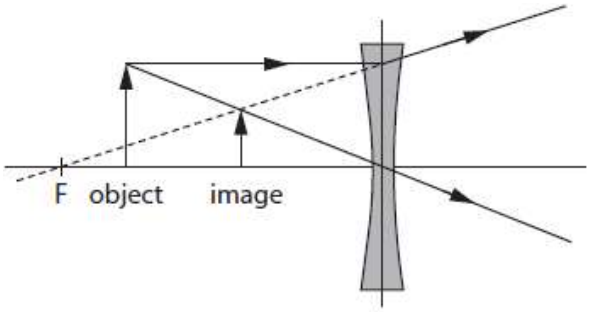

Figure 10

State what is meant by the term **virtual image**.

(1)

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.....

(Total for question = 1 mark)

Q11.

Figure 9 is a table of data for three coloured beams of light travelling from air into glass.

	wavelength in air	frequency in air	wavelength in glass	frequency in glass
red light	650 nm	4.6×10^{14} Hz	490 nm	4.6×10^{14} Hz
yellow light	580 nm	5.2×10^{14} Hz	440 nm	5.2×10^{14} Hz
blue light	470 nm	6.4×10^{14} Hz	350 nm	6.4×10^{14} Hz

Figure 9

Using data from Figure 9,

(i) state what happens to the wavelength of light when the light travels from glass into air

(1)

.....

.....

(ii) state what happens to the frequency of light when the light travels from glass into air.

(1)

.....

.....

(iii) The speed of blue light in air is 3.0×10^8 m/s.

Calculate the change in the speed of blue light when it travels from air into glass.
Use the equation

$$v = f \times \lambda$$

(3)

change in speed = ... m/s

(Total for question = 5 marks)

Mark Scheme

Q1.

Question Number:	Answer	Additional guidance	Mark
	a description to include: intensity reaches a peak value (1) additional information (1)	 other peaks irregular curve non-linear most of radiation outside visible spectrum	**(2)** AO 3 1a AO 3 1b

Q2.

Question number	Answer	Additional guidance	Marks
	A comparison that combines a maximum of four out of the following points to provide a reasoned response	Q does not ask for wavelengths specifically so colours will do	
	• both are sensitive about mid-range	yellow/green/blue	
	• peak at different wavelengths		
	• the insect will be better at long(er) wavelengths	red end	(4)
	• the insect will be much better at detecting shorter wavelengths	violet end	
	• the insect will be able to detect some wavelengths that humans cannot	ultraviolet	
	• both are insensitive at longer wavelengths	red end.	
	• the insect detects a greater range of wavelengths/colours than the human eye		
	• humans more sensitive than insects at 510 (nm)	accept 490 to 520	

Q3.

Question Number	Answer	Acceptable answers	Mark
(a)	**D** an ultraviolet wave		**(1)**

Question Number	Answer	Acceptable answers	Mark
(b)	Ultraviolet (from lamp) absorbed (by fluorescent substance/bank note) (1) (which) emits {visible/light} (into eye) (1)	Allow UV for ultraviolet Allow 'taken in' for absorbed Allow 'given out'/releases/fluoresces for emits 'Fluoresces' on its own is insufficient Mention of both ultraviolet AND visible/light only, scores 1 mark only	**(2)**

Q4.

	Answer	**Acceptable answers**	**Mark**
(a)	**light** → electrical → **chemical** energy energy energy (1) (1)	These answers must be in the correct order	**(2)**
(b)(i)	350 (J)	400 – 50 (J)	**(1)**
(b)(ii)	Substitution 50 ÷ 400 (1) or $50 \times 100/400$ (%) Evaluation 13(%) (1)	12.5(%), 0.125, 0.13 or 1/8 Give full marks for correct answer, no working	**(2)**
(c)(i)	An explanation linking the following points: black (1) (because) (good) absorber (of thermal radiation) (1)	absorbs / takes in heat radiation **ignore** references to: attract good emitter light dark / darker	**(2)**
(c)(ii)	an explanation linking any **three** of the following points: • (bag / water) absorbs thermal energy / heat / radiation (1) • (bag / water) radiates / emits thermal energy / heat	idea of energy input e.g. "sun heats the bag up" idea of energy output idea of more heat lost (to surroundings)at higher temperature "a bsorbing heat at	**(3)**

	/ radiation (1) • more heat radiated at higher temperature (1) • input and output are balanced (at steady temperature) (1)	same rate as radiating heat" (3) **ignore** (sun) light / rays	

Q5.

Question number	Answer	Mark
CS3	[x] A: wave velocity – decreases; wavelength – decreases B is not correct because the wavelength does not increase C and are not correct because the wave velocity does not increase	(1) AO1

Q6.

	Answer	Acceptable answers	Mark
(a)	A		**(1)**
(b)	alpha particles (In the left section) gamma rays (centre section) infrared radiation (right section) (2)	Any one in correct position for one mark, all three in correct position for two marks	**(2)**
(c)	C		**(1)**
(d)	A description to include The purpose of using gamma radiation (1) Some relevant detail about how it achieves the purpose (1)	Purposes may include sterilising food /medical equipment detection / treatment of cancer imaging /detect flaws in materials	**(2)**

		Indicative Content	Mark
QWC	***(e)**	An explanation including some of the following points	**(6)**

		Results obtained: • Herschel: temperature on thermometer • Ritter: speed of darkening of silver chloride paper Trend of results: • Herschel: hotter towards red end • Ritter: quicker towards blue/violet end Extension of experiment to get more results: • Herschel: measure below red; found it even hotter • Ritter: measure above blue/violet; paper darkened quicker Conclusion: • Herschel: Must be radiation below red (Infra Red) • Ritter: Must be radiation above blue/violet (UV)	
Level	**0**	No rewardable content	
1	**1 - 2**	• A limited description of either some results or conclusions from either experiment. For example: They measured temperature across the spectrum and found that temperature changed. They put silver chloride paper in the spectrum and found that it darkened at different speeds with different colours. The answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3 - 4**	• A simple explanation of results and conclusions from both experiments . For example: Herschel measured the temperature across the spectrum and found it hotter towards the red end. This was infra red radiation. Ritter measured the darkening of chloride paper across the spectrum. It was quicker towards the violet end. They had discovered ultra violet. • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately	

		• spelling, punctuation and grammar are used with some accuracy
3	**5 - 6**	• a detailed explanation of all the results obtained from both experiments and the conclusions from these results. For example a response as for level 2 given above but with detail about results being obtained from outside the visible spectrum • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors

Q7.

Question Number	Answer	Additional guidance	Mark
(i)	tangent drawn between 20 and 40 ms (1) selection of suitable values and substitution from their tangent (1) e.g. 130 ÷ 60 evaluation (1) 2.0 unit (1) °C/ms	 range 1.8-2.5 2000 °C/s award full marks for correct answer with no working	**(4)** AO 1 1 AO 2 1

Question Number	Answer	Additional guidance	Mark
(ii)	an explanation linking:		(3) AO 3 2a AO 3 2b
	the temp rises because the (rate of) energy supplied is greater than the (rate of) energy being radiated. (1)	accept 'absorbed/in' for supplied. energy in greater than energy out lamp radiates more as temperature increases	
	at constant temperature the lamp radiates energy at the same **rate** at which it is being supplied (2)	lamp radiates at the same **power** at which it is being supplied for 2 marks **power** out = **power** in award 1 mark for at constant temperature the energy supplied = energy radiated	

Q8.

Question number	Answer	Additional guidance	Marks
	An explanation that combines identification via a judgement (1 mark) to reach a conclusion via justification (1 mark):		**(2)**
	• anomalous result / it is at (angle of incidence) 40° (1)	the 4th cross the one in the middle. may be identified on the graph	
	• (because) it is further away from the curve of best fit than the other results (1)	does not follow same pattern as the others / not in line etc. explains the cause of the error e.g. misreading protractor suggests a plausible correct result such as 40(°),25(°) ignore points at i = 40(°) and 50(°) are the same	

Q9.

Question Number	Answer	Additional guidance	Mark
	suggestion to include **one** from (ultraviolet/UV) is (the most) harmful to the eyes (1) protects eyes from damage/harm (from UV rays) (1)	 (UV) can damage eyes protects against cataracts/cancer accept makes it more comfortable in bright sunlight	**(1)**

Q10.

Question	Answer	Additional guidance	Mark
	one from the image/ it cannot be produced /projected (on a screen) (1) the rays do not actually pass through the image (1) image formed when imaginary rays meet (1) the image is on the same side of the lens as the object (1) the image is upright (1)	ignore image is not real allow the image is in front of the lens not inverted / up side down	**1** **AO1.1**

Q11.

	Answer	Additional guidance	Mark
(i)	(wavelength) increases (from glass to air)	gets longer/gets bigger allow quoted values allow (wavelength) decreases from air to glass	(1) AO3.1

	Answer	Additional guidance	Mark
(ii)	(frequency) doesn't change	remains constant / stays the same / nothing allow quoted values	(1) AO3.1

	Answer	Additional guidance	Mark
(iii)	substitution (1) (speed in glass =) 6.4 (× 10^{14}) × 350 (× 10^{-9}) evaluation of speed in glass (1) 2.2 × 10^{8} (m/s) subtraction involving 3 × 10^{8} (m/s) (1) ALTERNATIVE METHOD calculation of change in wavelength for blue light (1) 470(× 10^{-9}) - 350 (× 10^{-9}) OR 120 (× 10^{-9}) substitution (1) (difference in speed =) 6.4 (× 10^{14}) x 120 (× 10^{-9}) substitution (1) (difference in speed =) 6.4 (× 10^{14}) x 120 (× 10^{-9}) evaluation (1) 7.7 × 10^{7} (m/s)	allow numbers that round to 2.2 × 10^{8} e.g. 2.24 × 10^{8} independent mark 7.6 × 10^{7} (m/s) scores 3 marks ecf their change in wavelength for this mark only ecf their change in wavelength for this mark only award full marks for the correct answer without working	**(3)** **AO3.2**